AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 1 · STUDENT LESSON

The ML Lifecycle

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

AI-2 Week 11: training, selection and final evaluation use different evidence.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Distinguish fitting a rule from applying a fixed rule. Try the recall before opening notes.

Your learning target

I can distinguish **fitting** a rule from applying a fixed rule.

I can trace a small learning lifecycle with separate data.

I can report test **evidence** with its limits.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Creator Defense. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Machine learning is an **evidence** cycle: define a **prediction** problem, prepare examples, fit a model, evaluate on unseen data, and improve one diagnosed cause.



Your target

LEARNING TARGET A team **label**s photos, fits a model and celebrates one correct new photo. List the missing **evaluation** steps and define the intended use before testing.

Builder name and experiment date:

Readiness for the connected explanation

> A fixed rule matches 3 **label**s out of 4 tested cases. What fraction is correct, and does that prove it works on all future cases?

This week's end goal

By the end, I can explain the ml lifecycle using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied The ML Lifecycle case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use the ml lifecycle to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or **evaluation** decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

In Grade 7 you programmed an activity chooser using a rule supplied by a person. This week you will choose a rule by comparing its **prediction**s with **label**led examples.

This small supervised learning task makes the learning **process** visible: people define the task and possible rules; **training** data determines which candidate fits best.

Imagine fictional practice cards with a measured width x and a category **label**, 0 or 1. Width uses invented units. We want to predict the category of another card from its width.

The **label**s are provided **observation**s in this exercise, not values generated by our **prediction** code. These invented cards demonstrate a method; they provide no **evidence** about real objects or students.

Purpose	Width x	Observed label
Training	1	0
Training	3	0
Training	5	1
Training	7	1
Held out for later testing	2	0
Held out for later testing	6	1

Before **fitting**, set aside the final two cards. They are visible here so you can follow the complete example, but do not use them to choose a rule. Our candidates predict 1 when width is at least 2, 4 or 6, and 0 otherwise.

Choose the candidate with the fewest **training** mistakes. If candidates tie, choose the smaller threshold; fix this rule before testing.

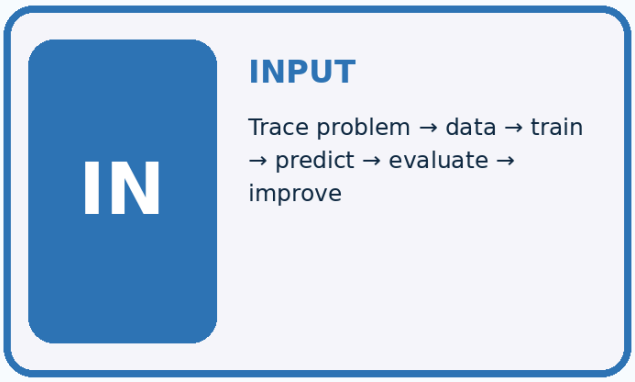
See how it works

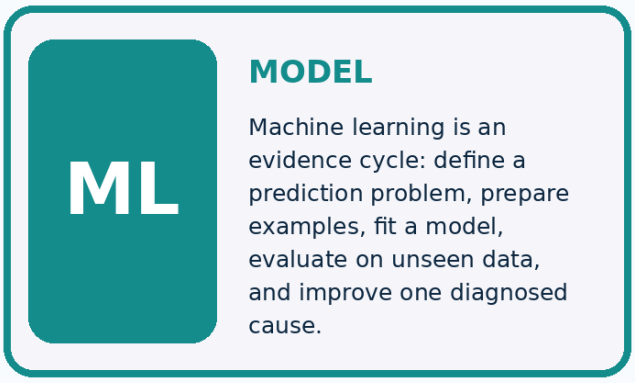
> Follow the mechanism for The ML Lifecycle. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and **evaluation** proof separately.







Question to explain

QUESTION Trace problem → data → train → predict → evaluate → improve, and identify where human judgment enters.

MODEL IDEA **Machine learning** is an **evidence** cycle: define a **prediction** problem, prepare examples, fit a model, evaluate on unseen data, and improve one diagnosed cause.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Goal: classify a leaf as healthy or spotted for a classroom plant helper. Prepare **label**ed leaf examples and reserve unseen test examples.

Draw a fresh data → model → **prediction** → **evaluation** trace:

A closer explanation

```
train = [(1, 0), (3, 0), (5, 1), (7, 1)]
# Predict 1 when x is at least the threshold; otherwise 0.
def predict(x, threshold):
    return int(x >= threshold)

for threshold in [2, 4, 6]:
    errors = 0
```

```
    for x, label in train:
        if predict(x, threshold) != label:
            errors = errors + 1
    print(threshold, errors)
```

int(True) is 1 and int(False) is 0. For threshold 2, the **training prediction**s are 0, 1, 1, 1: the width 3 card is wrong. For threshold 4, they are 0, 0, 1, 1: none is wrong.

For threshold 6, they are 0, 0, 0, 1: the width 5 card is wrong. The printed rows are 2 1, then 4 0, then 6 1. Select threshold 4 using this **training evidence** and record it.

The loop reports errors; the learner makes and records the selection.

Keep fitting and testing separate

STAGE	INFORMATION USED	RECORD
Define and split	Task and candidate rules	Training and test cards
Fit and freeze	Training labels only	Threshold 4 selected
Test and report	Two held out cards	2 correct out of 2

Read down the stages. Test results may inform later development, but they must not silently change the rule being tested.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

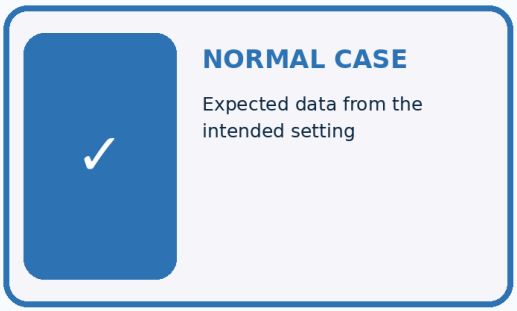
Compare the cases

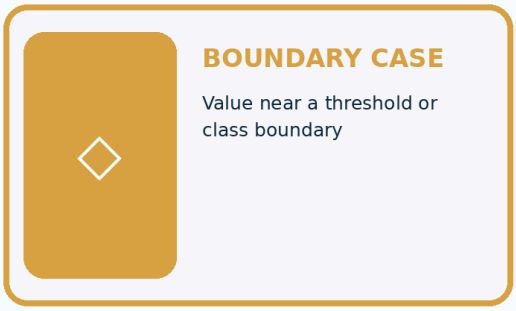
> Compare the supplied cases for The ML Lifecycle. Identify the detail that changes the conclusion.

EVALUATION DESIGN

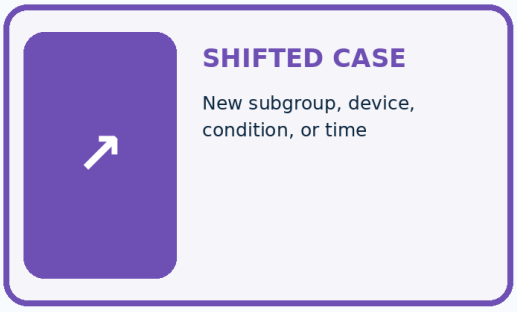
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous **label**, or uncertain **prediction**.

MISSING Necessary field, **label**, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: The fresh width-6 leaf is actually healthy.

Predicted result

Prediction: The frozen model is correct on 1/2 test leaves.

Why the result follows

Reason: The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of The ML Lifecycle to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.


1
MACHINE LEARNING
fit a model from examples
2
TRAINING
adjust model parameters using examples
3
INFERENCE
apply the fitted model to new input
4
EVALUATION
measure behavior on unseen examples

Term	Precise meaning	My ML example
machine learning	fit a model from examples	
training	adjust model parameters using examples	
inference	apply the fitted model to new input	
evaluation	measure behavior on unseen examples	

MISCONCEPTION TO REPAIR **Training** once and seeing one correct **prediction** proves the model works.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Feature	An input value used to make a prediction
Label	An observed target value supplied for an example
Training	Using examples to select or adjust a model
Held out data	Examples reserved from fitting and model choices for evaluation

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for The ML Lifecycle, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 1 • WORKED EVIDENCE

1. Goal: classify a leaf as healthy or spotted for a classroom plant helper.
2. Prepare labeled leaf examples and reserve unseen test examples.
3. Fit the model on training data; predict labels for the test set.
4. Measure errors, diagnose a cause, change one factor, and retest.

Step 1

STEP 1 Goal: classify a leaf as healthy or spotted for a classroom plant helper.

Step 2

STEP 2 Prepare **label**ed leaf examples and reserve unseen test examples.

Step 3

STEP 3 Fit the model on **training** data; predict **label**s for the test set.

Step 4

STEP 4 Measure errors, diagnose a cause, change one factor, and retest.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Worked explanation · The ML Lifecycle

Freeze means do not change the chosen threshold during this **evaluation**. With threshold 4, widths 2 and 6 receive **prediction**s 0 and 1. Compare them with the held out **label**s: both match. Report 2 correct out of 2 held out cards.

Training results and test results answer different questions: did the rule fit the examples used to select it, and how did that fixed rule perform on these separate examples?

Keep threshold 4 fixed. An additional card has width 4 and observed label 0. Predict its category, decide whether it is correct, and update the held out count from 2 of 2.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: The ML Lifecycle

Follow one complete plant-classifier lifecycle without tuning on the final test.

The facts we will use

Supplied facts

Toy **training** leaves have widths 1 healthy, 3 healthy, 5 spotted, 7 spotted. Candidates predict spotted at width>=2 or >=4.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/1

The ML Lifecycle

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit using training

Cutoff 2 makes one error at width 3; cutoff 4 makes zero training errors.



2. Freeze candidate

Choose cutoff 4 under the stated training-error rule.



3. Test once

Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

CHANGE ONE THING

The fresh width-6 leaf is actually healthy.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit using training	Cutoff 2 makes one error at width 3; cutoff 4 makes zero training errors.
2. Freeze candidate	Choose cutoff 4 under the stated training-error rule.
3. Test once	Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Two invented test leaves cannot establish useful real-world plant classification.

Keep the stated limit

Read the **label**led picture: Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: The fresh width-6 leaf is actually healthy. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the The ML Lifecycle claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Training once and seeing one correct prediction proves the model works. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE **Training** once and seeing one correct **prediction** proves the model works.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

This threshold model learned only a choice among three candidate values. People supplied the **feature**, **label**s, candidate family and tie rule.

Learning does not imply understanding, and perfect **training** performance does not guarantee good **prediction**s. Two correct test cards are too few to establish reliable performance for a broader population.

Even a larger score would depend on how examples were collected and what use was intended.

Your report says the fixed model matched 2 of 2 held out cards. Write one supported claim, one claim this does not support, and one sensible next step.

STUDENT LESSON · 8 OF 18

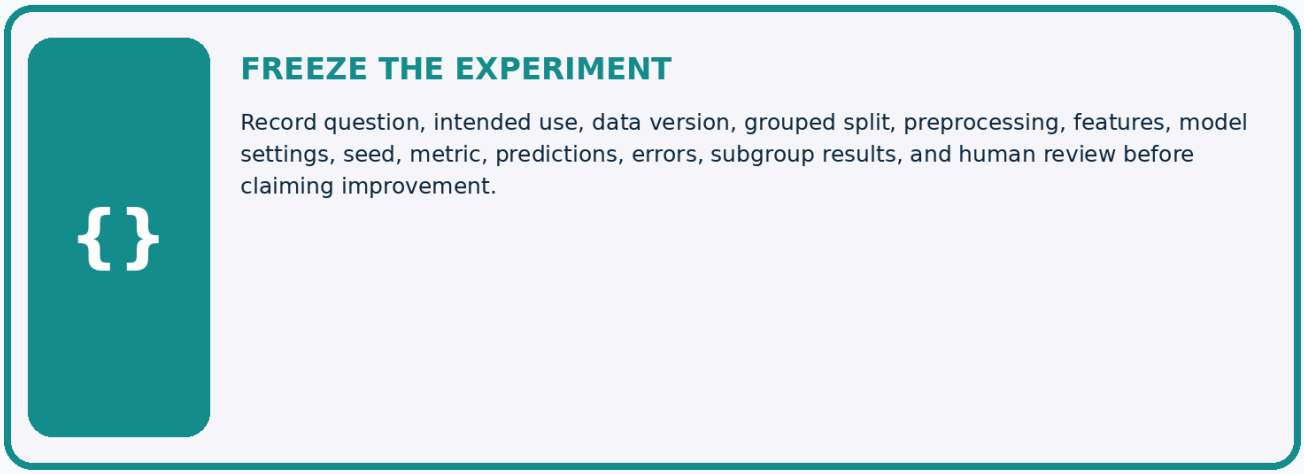
Practise together

Practise The ML Lifecycle with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and **evaluation** rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + **feature**s + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

A learner sees the width 4 test error, changes the threshold to 5, and reports the same cards as a fresh final test. The test **label**s have now influenced the model. Keep the original 2 of 3 result for the original rule.

Record the change as development, then reserve genuinely unused examples for a later **evaluation**. Do not delete the difficult card or rename reused cards as new **evidence**.

A lifecycle continues after **fitting**: define the intended use, prepare and separate data, fit, evaluate, review the limits, and monitor any approved use. **Evaluation** can send us back to development.

That return needs an honest record and a new final **evaluation**; it does not erase the earlier error.

Practise with less help: Trace one candidate

Use workbook W2 for the connected example “The **machine learning** lifecycle”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for The **machine learning** lifecycle. Proposed explanation: “The model wrote its own task”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

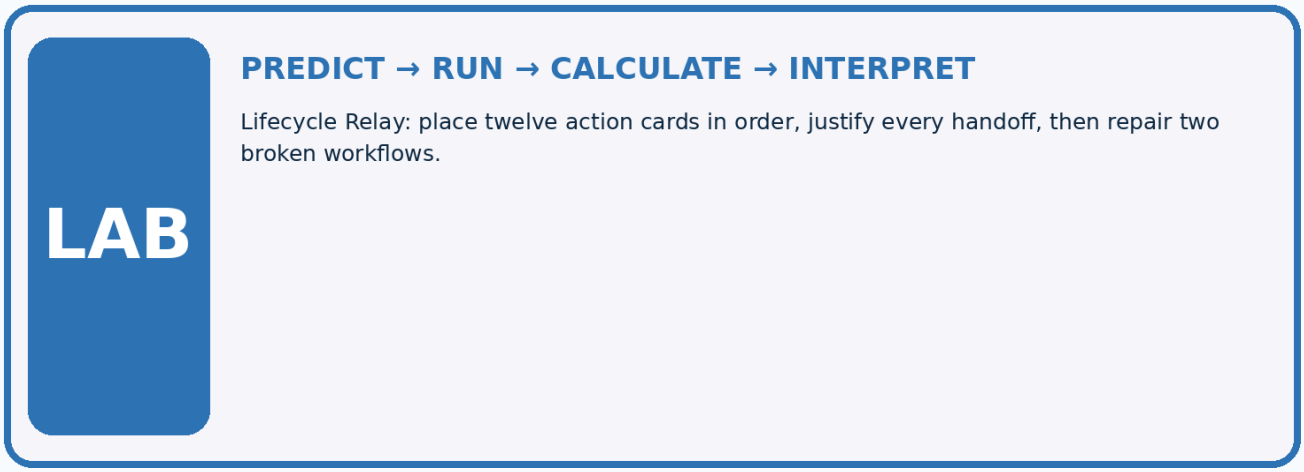
Run the investigation

> Investigate The ML Lifecycle. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Lifecycle Relay: place twelve action cards in order, justify every handoff, then repair two

broken workflows.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal
Prediction: _____
Actual: _____
Calculation / evidence: _____
Meaning / next test: _____

Case: Boundary
Prediction: _____
Actual: _____
Calculation / evidence: _____
Meaning / next test: _____

Case: Missing
Prediction: _____
Actual: _____
Calculation / evidence: _____

> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

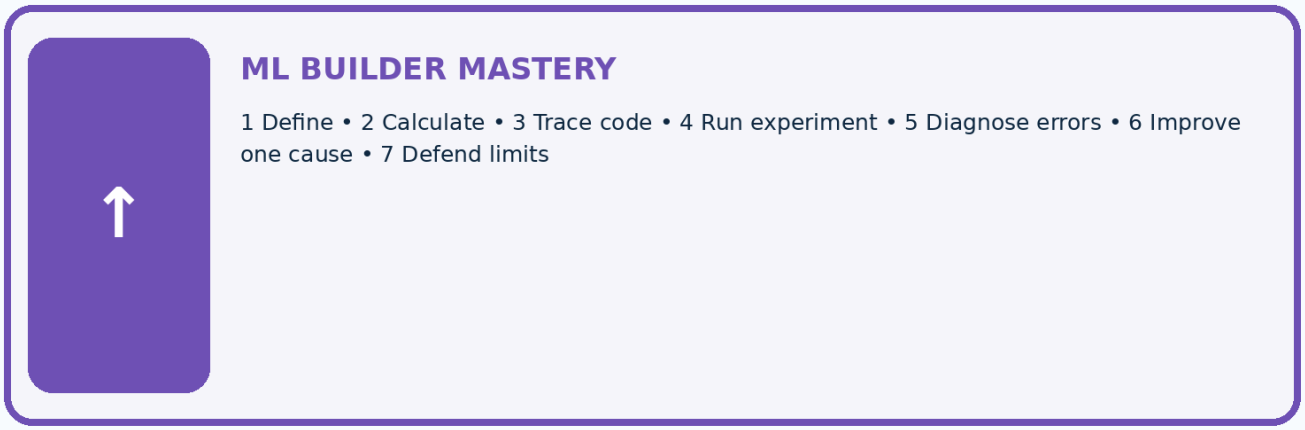
Build your understanding

> Move from recall to tracing, diagnosis and transfer in The ML Lifecycle. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **machine learning**, **training**, **inference**, **evaluation**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE **Training** once and seeing one correct **prediction** proves the model works.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is

completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week’s idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Write the Responsible Plant Classifier problem statement, intended user, **prediction** target, success measure, and human-review point.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

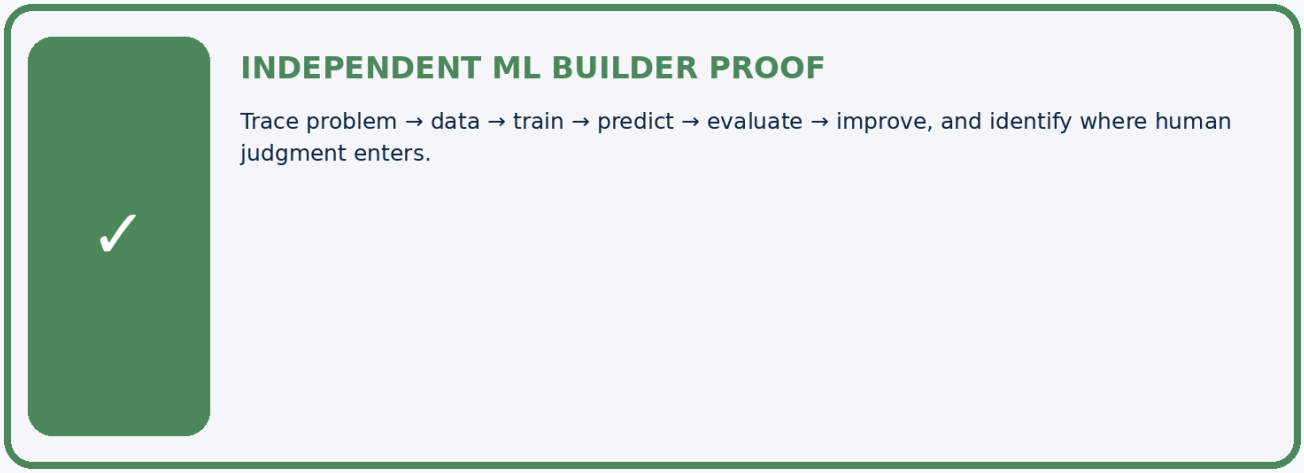
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Trace problem → data → train → predict → evaluate → improve, and identify where human judgment enters.

- ☐ Define the task, **feature**/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.

> ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: **Feature**s, **Label**s, Targets.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete The ML Lifecycle case for supported practice. Repeated examples do not become fresh assessments.

Week 1 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A team **label**s leaf photos, trains a healthy/spotted classifier and shows one correct **prediction**.

The question

Order the lifecycle and identify missing **evidence**.

Worked explanation

Define use and **label**s, collect data, separate **evaluation** data, train, predict, evaluate and improve. One correct result is insufficient. Test unseen varied cases and retain human responsibility.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Evidence drives the ML lifecycle

Evidence or stage	Value or result
Define	Intended use and labels
Prepare	Collect and separate evaluation data
Train	Fit on training data
Evaluate	Inspect unseen results
Revise	Change a cause and retest

A team **label**s leaf photos, trains a healthy/spotted classifier and shows one correct **prediction**.

Order the lifecycle and identify missing **evidence**.

Define use and **label**s, collect data, separate **evaluation** data, train, predict, evaluate and improve. One correct result is insufficient. Test unseen varied cases and retain human responsibility.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Fit and evaluate independently

New **training** cards are (width 2, **label** 0), (4, 0), (6, 1), (8, 1). Candidates are 3, 5 and 7. Predict 1 when width is at least the threshold.

Count **training** mistakes, select the fewest with smaller threshold breaking ties, and record the choice before testing. Held out cards are (3, 0), (7, 1), (5, 0).

Give each test **prediction**, the correct count and a limited conclusion.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 1 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A team selects a model before deciding what outcome to predict or where the data comes from. Name two missing lifecycle decisions and their effect.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Write the Responsible Plant Classifier problem statement, intended user, **prediction** target, success measure, and human-review point.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A team **label**s photos, fits a model and celebrates one correct new photo. List the missing **evaluation** steps and define the intended use before testing.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate The ML Lifecycle, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Follow one complete plant-classifier lifecycle without tuning on the final test.

By the end, I can	How I will show it
Distinguish fitting a rule from applying a fixed rule	Explain the deciding step, show a trace or test, and state one limit.
Trace a small learning lifecycle with separate data	Explain the deciding step, show a trace or test, and state one limit.
Report test evidence with its limits	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain The ML Lifecycle without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Distinguish **fitting** a rule from applying a fixed rule

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Follow one complete plant-classifier lifecycle without tuning on the final test.

Practice 1: explain the deciding evidence

Which **label**led step matches this detail? Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

1. Test once

2. Freeze candidate

3. Fit using **training**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Test once Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. Read the picture in order.

Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The fresh width-6 leaf is actually healthy. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The frozen model is correct on 1/2 test leaves.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The frozen model is correct on 1/2 test leaves. The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

Repair a missing building block

Review Creator Defense (ai-3/36). Goal: Explain the **mechanism** and terminology of generative systems defense. Short practice: Which numbers would you display to make the defense honest?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, **label** your result as a trace rather than an execution.

Keep your completed work

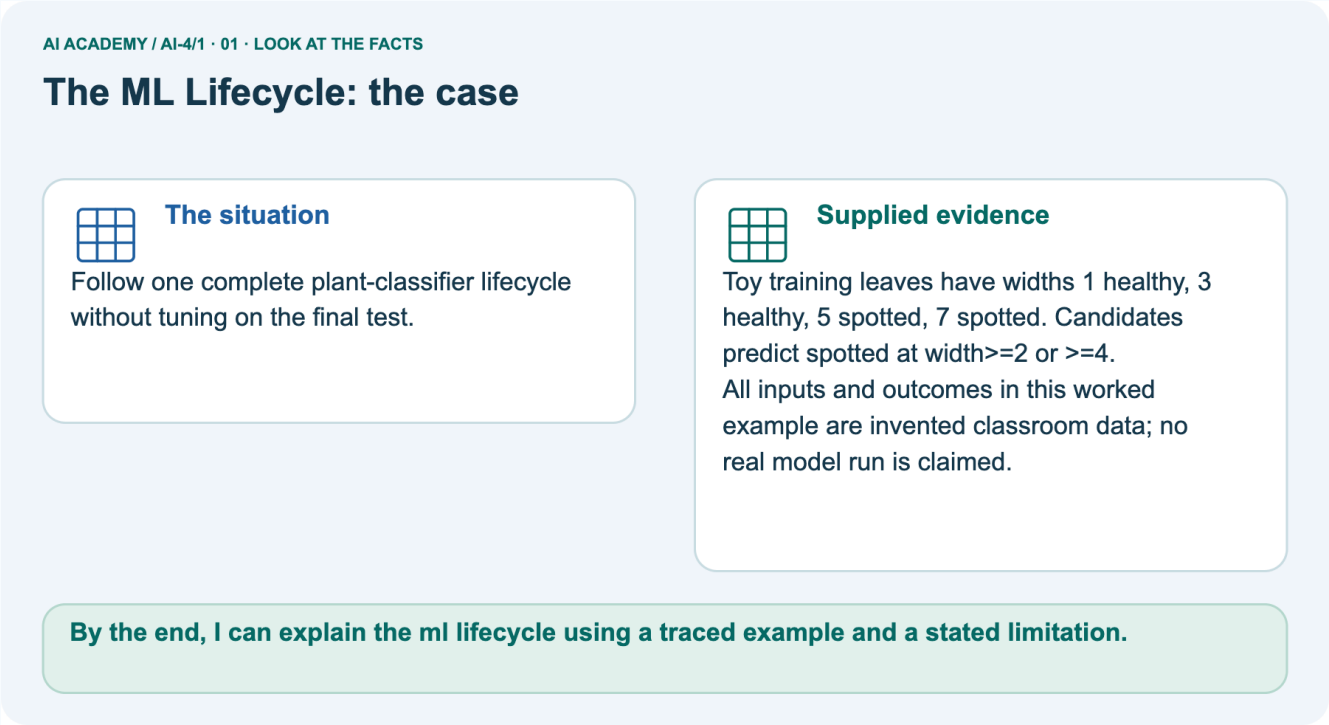
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each The ML Lifecycle graphic using its labels, deciding evidence and relevant limitation.

01. The ML Lifecycle: the case



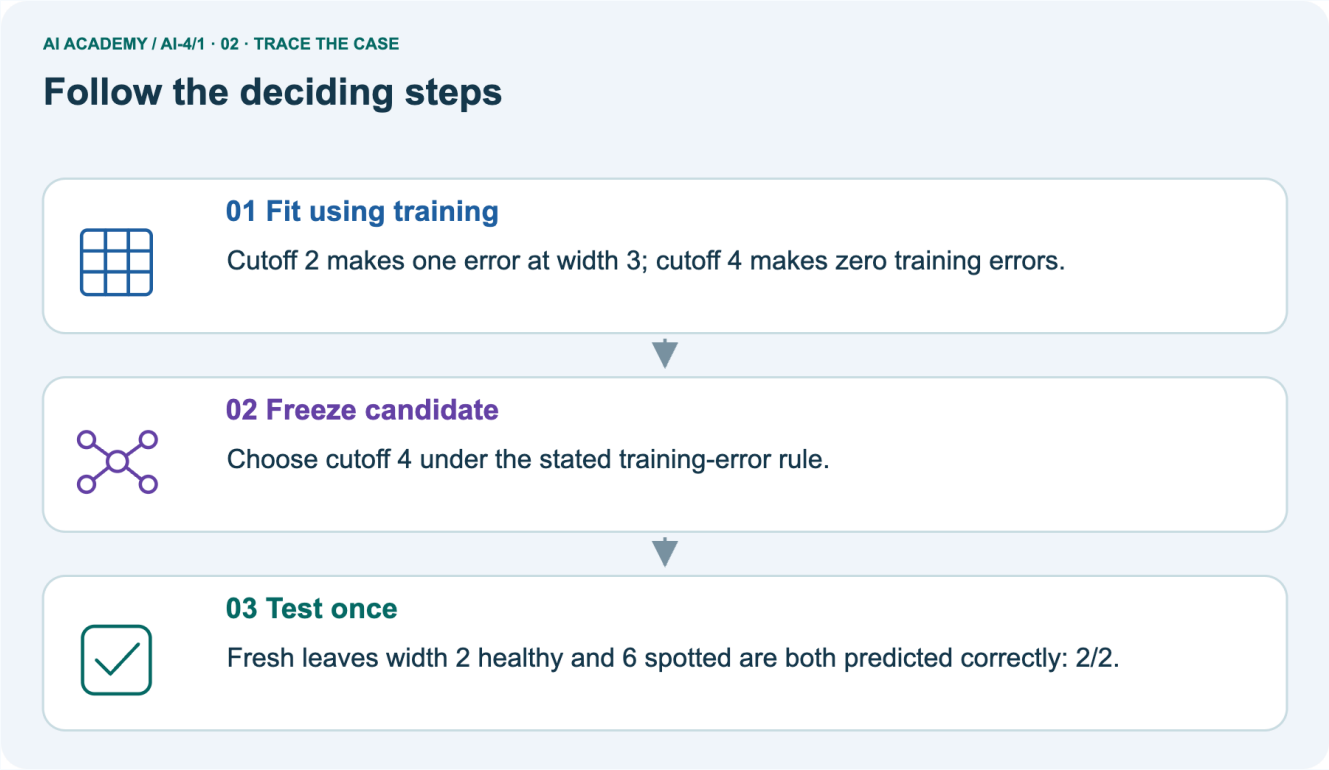
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow one complete plant-classifier lifecycle without tuning on the final test. Supplied facts: Toy **training** leaves have widths 1 healthy, 3 healthy, 5 spotted, 7 spotted. Candidates predict spotted at width>=2 or >=4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

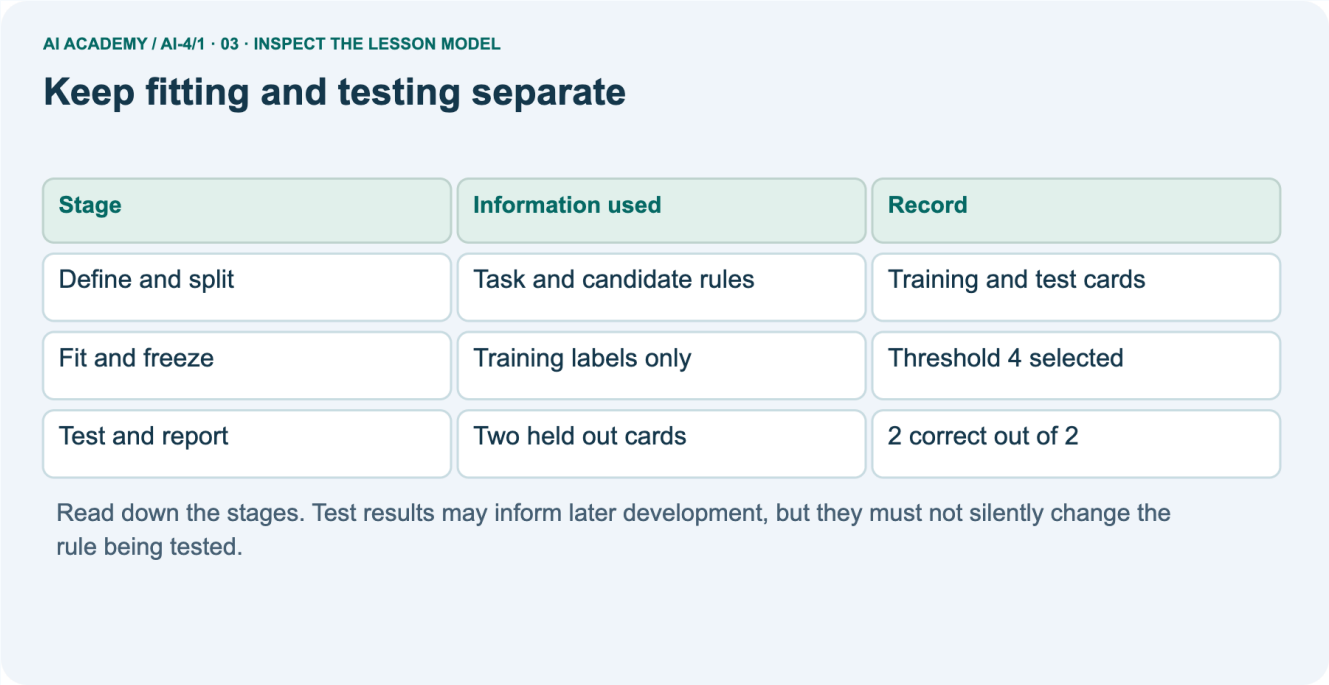


Follow the labelled picture with the matching explanation. Open the vector diagram

Fit using **training**: Cutoff 2 makes one error at width 3; cutoff 4 makes zero **training** errors. → Freeze candidate: Choose cutoff 4 under the stated **training**-error rule. → Test once: Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Keep fitting and testing separate

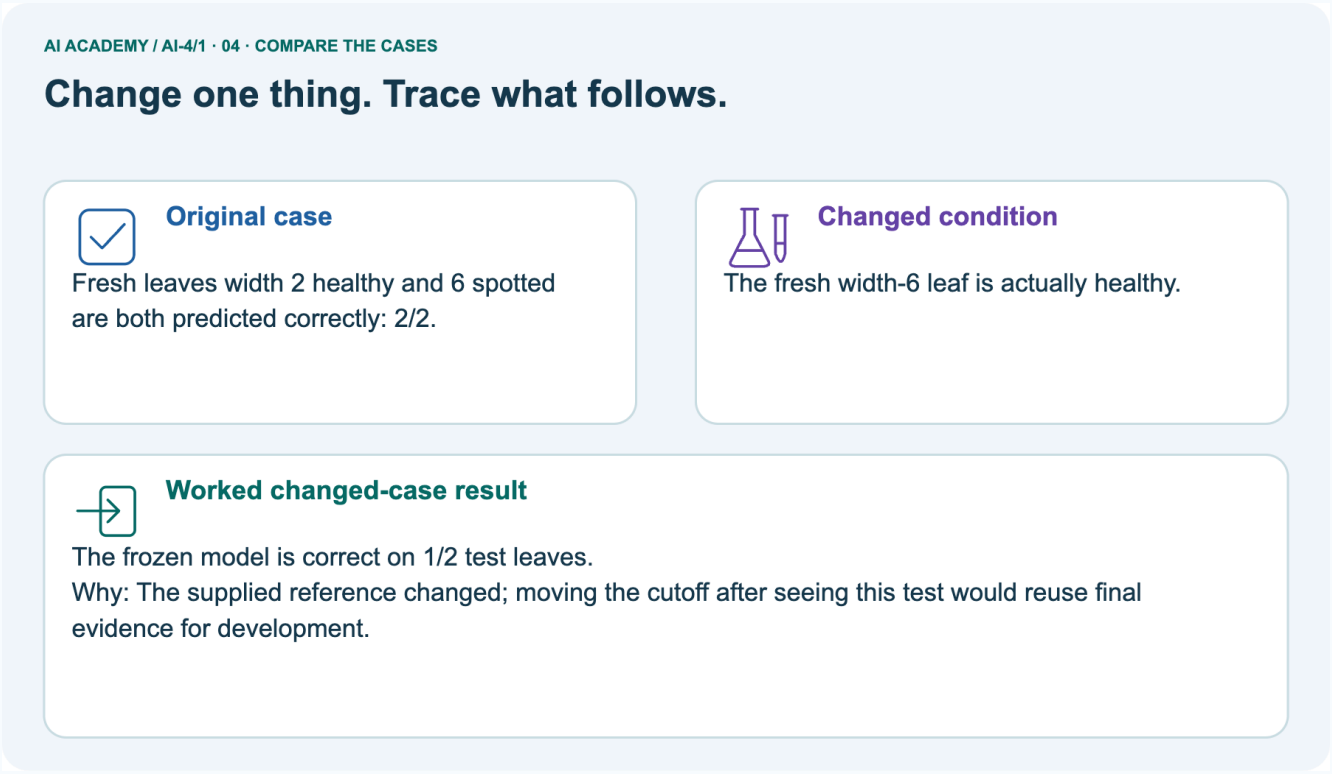


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep **fitting** and testing separate. Define and split | Task and candidate rules | **Training** and test cards. Fit and freeze | **Training label**s only | Threshold 4 selected. Test and report | Two held out cards | 2 correct out of 2

> Read the **label**s and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



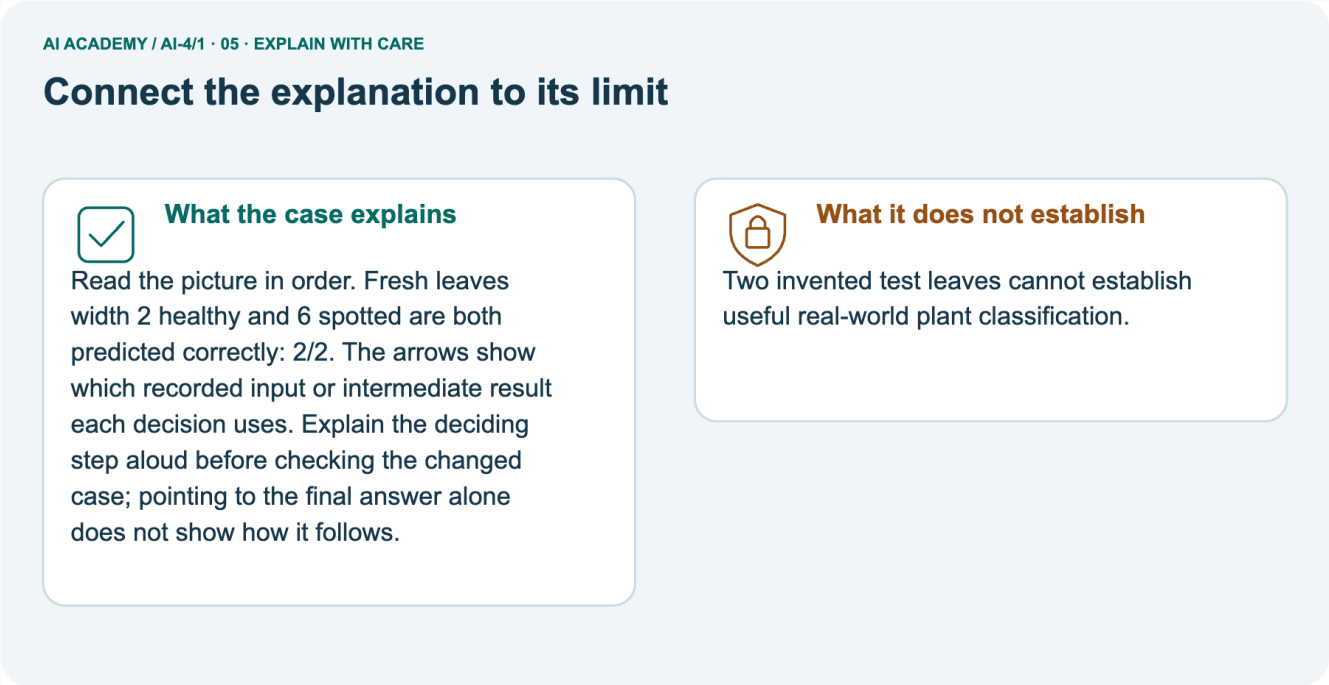
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. Changed condition: The fresh width-6 leaf is actually healthy. Expected result: The frozen model is correct on 1/2 test leaves.

Reason: The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



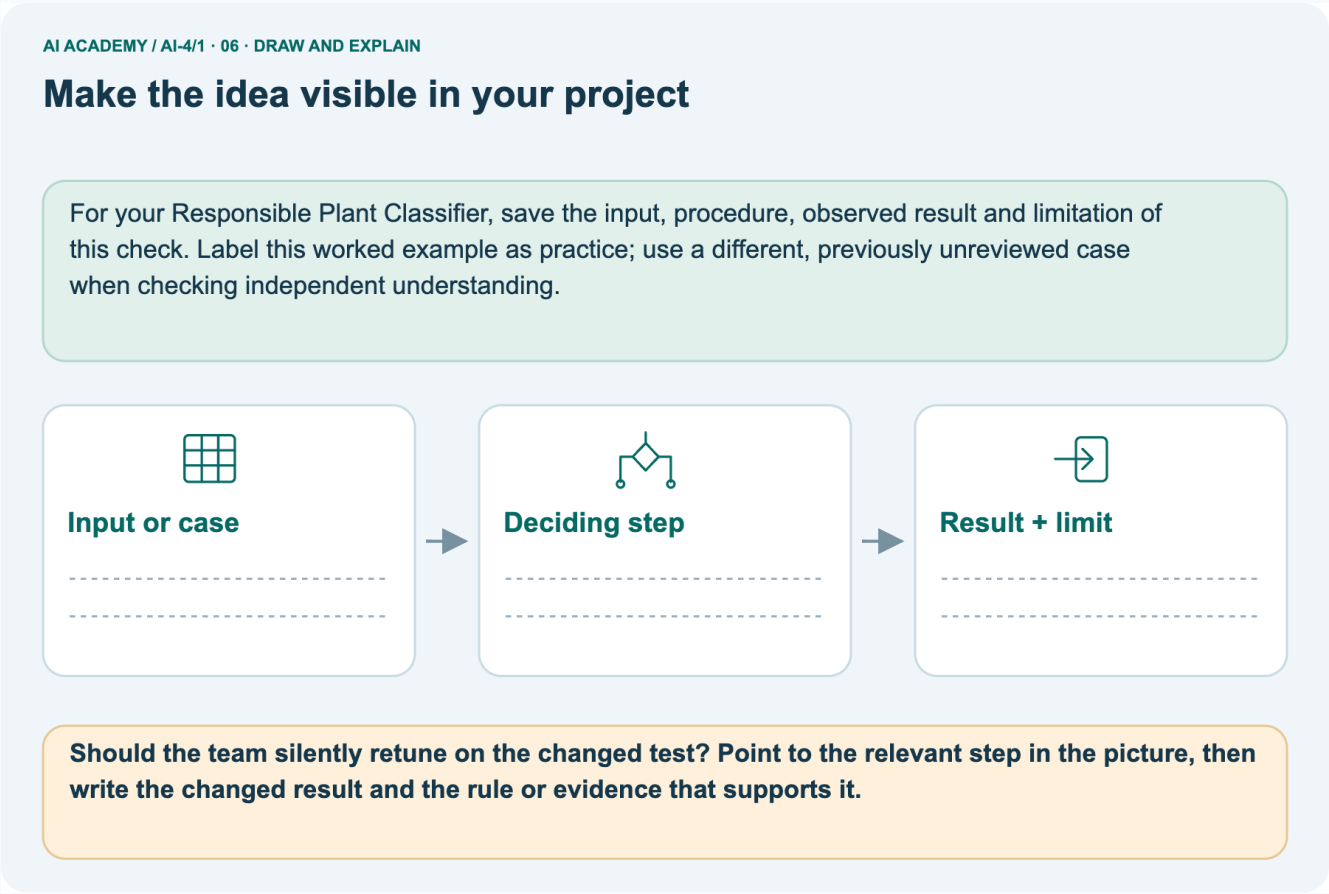
Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Two invented test leaves cannot establish useful real-world plant classification.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.